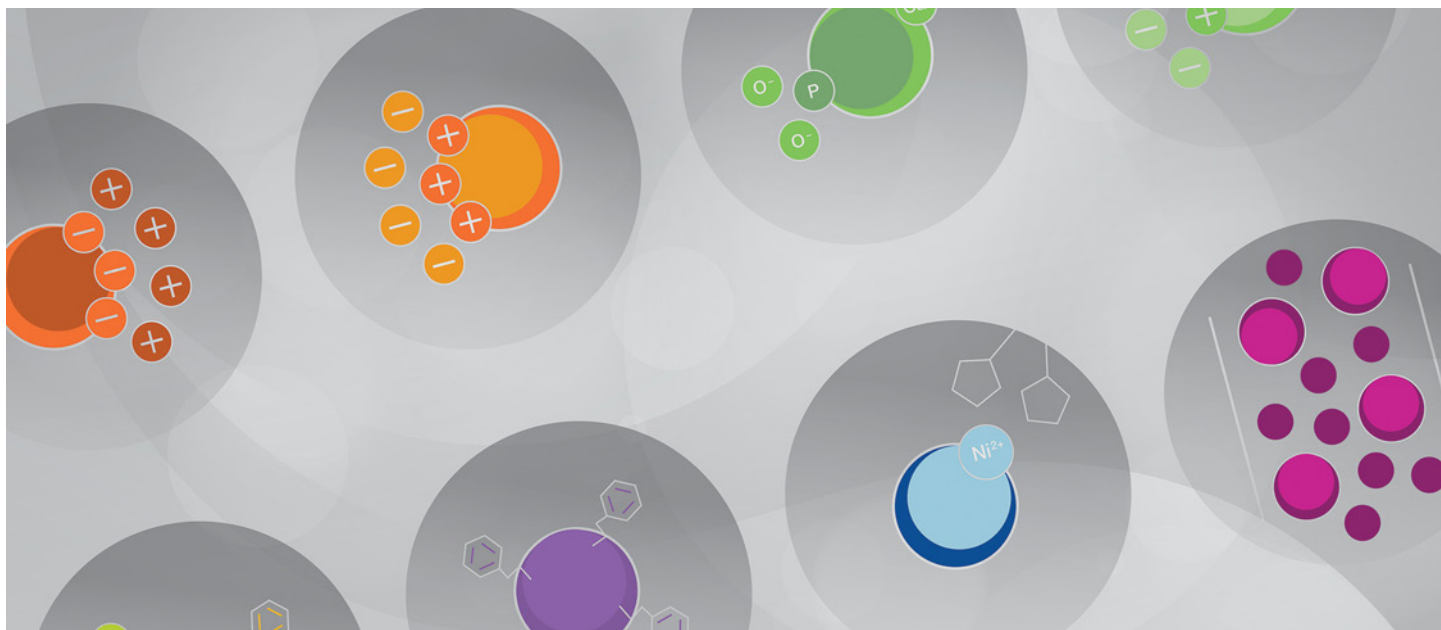




MIXED-MODE

Mixed-Mode Resins/Media Selection Guide

Bio-Rad offers a wide range of mixed-mode resins/media, which can be used for purification of antibodies, proteins, and viruses as well as removal of aggregates, endotoxins, host cell proteins, and DNA.

Use this guide to select the optimal resin/media for your purification project.

Optimal Mixed-Mode Resins/Media for Multiple Purification Applications

Property	Process Purification Resins/Media				
Resin/media type	Nuvia cPrime	Nuvia aPrime 4A	CHT™ Ceramic Hydroxyapatite Type I	CHT Ceramic Hydroxyapatite Type II	CHT Ceramic Hydroxyapatite XT
Chromatography type	Mixed-mode (HIC and cation exchange)	Mixed-mode (HIC and anion exchange)	Mixed-mode (cation exchange and calcium affinity)	Mixed-mode (cation exchange and calcium affinity)	Mixed-mode (cation exchange and calcium affinity)
Particle size	70 ± 10 µm	50 ± 10 µm	20 ± 2, 40 ± 4, 80 ± 8 µm	20 ± 2, 40 ± 4, 80 ± 8 µm	40 ± 4 µm
Ionic capacity	123 ± 20 µeq/ml	100 ± 20 µeq/ml	—	—	—
Dynamic binding capacity (DBC)	>40 mg hIgG/ml >60 mg lactoferrin/ml	≥50 mg/ml mAb*	25–60 mg IgG/ml at 300 cm/hr**	15–25 mg IgG/ml at 300 cm/hr**	>60 mg/ml mAb (pI 9.2)***
Recommended linear flow rate	50–600 cm/hr	50–300 cm/hr	50–300 cm/hr	50–300 cm/hr	50–300 cm/hr
pH stability	Short-term: 3–14 Long-term: 4–13	Short-term: 2–14	6.5–14	6.5–14	6.5–14

HIC, hydrophobic interaction chromatography; hIgG, human immunoglobulin G; mAb, monoclonal antibody.

* 10% breakthrough capacity determined with 1.0 mg/ml of an acidic mAb (pI ~6.9) in 20 mM sodium phosphate, pH 7.8.

** 40 µm particles, 300 cm/hr, 5 mM sodium phosphate, 25 mM sodium chloride, pH 6.5.

*** At 100 cm/hr with 5 mM sodium phosphate, 25 mM sodium chloride, pH 7.

To see how these mixed-mode resins/media have been used in various purification applications, refer to the following bulletins:

- mAb purification: [6875](#)
- IgG purification: [6793](#)
- Adenovirus purification: [6807](#)
- Mammalian virus purification: [6790](#)
- Protein purification: [6810](#)
- Aggregate removal: [6808](#)
- Endotoxin removal: [6813](#)
- Host DNA removal: [6881](#)

Ordering Information

Nuvia cPrime Resin	Catalog #	Specification
Foresight™ Nuvia cPrime Plates	732-4705	2 x 96-well, 20 µl
Foresight Nuvia cPrime	732-4807	200 µl
RoboColumn Unit	732-4808	600 µl
Foresight Nuvia cPrime Column	732-4722	1 ml
	732-4742	5 ml
Nuvia cPrime Resin	1563401	25 ml in 20% ethanol
	12003716	25 ml in 2% benzyl alcohol
	1563402	100 ml in 20% ethanol
	12003717	100 ml in 2% benzyl alcohol
	156-3403	500 ml in 20% ethanol
	12003718	500 ml in 2% benzyl alcohol
	156-3404	1 L in 20% ethanol
	12003719	1 L in 2% benzyl alcohol
	156-3405	5 L in 20% ethanol
	12003720	5 L in 2% benzyl alcohol
	156-3406	10 L in 20% ethanol
	12003721	10 L in 2% benzyl alcohol

Nuvia aPrime 4A Resin	Catalog #	Specification
Foresight Nuvia aPrime 4A Plates	12007411	2 x 96-well, 20 µl
Foresight Nuvia aPrime 4A	12007394	200 µl
RoboColumn Unit	12007395	600 µl
Foresight Nuvia aPrime 4A	12007392	1 ml
Column	12007393	5 ml
Nuvia aPrime 4A Resin	12007397	25 ml in 20% ethanol
	12007396	100 ml in 20% ethanol
	12007379	500 ml in 20% ethanol
	12007380	5 L in 20% ethanol
	12007391	10 L in 20% ethanol

CHT Ceramic Hydroxyapatite Type I Media	Catalog #	Particle Size	Specification
Foresight CHT Type I Plates	732-4716	40 µm	2 x 96-well, 20 µl
Foresight CHT Type I	732-4822	40 µm	200 µl
RoboColumn Unit	732-4823	40 µm	600 µl
Foresight CHT Type I Column	732-4735	40 µm	1 ml
	732-4755	40 µm	5 ml
CHT Type I Media	1582000	20 µm	10 g
	1570020	20 µm	100 g
	157-0021	20 µm	1 kg
	1584000	40 µm	10 g
	1570040	40 µm	100 g
	157-0041	40 µm	1 kg
	157-0045	40 µm	5 kg
	1588000	80 µm	10 g
	1570080	80 µm	100 g
	157-0081	80 µm	1 kg
	157-0085	80 µm	5 kg

CHT Ceramic Hydroxyapatite Type II Media	Catalog #	Particle Size	Specification
Foresight CHT Type II Plates	732-4718	40 µm	2 x 96-well, 20 µl
Foresight CHT Type II	732-4825	40 µm	200 µl
RoboColumn Unit	732-4826	40 µm	600 µl
Foresight CHT Type II Column	732-4736	40 µm	1 ml
	732-4756	40 µm	5 ml
CHT Type II Media	1582200	20 µm	10 g
	1572200	20 µm	100 g
	157-2100	20 µm	1 kg
	1584200	40 µm	10 g
	1574000	40 µm	100 g
	157-4100	40 µm	1 kg
	157-4500	40 µm	5 kg
	1588200	80 µm	10 g
	1578000	80 µm	100 g
	157-8100	80 µm	1 kg
	157-8500	80 µm	5 kg

CHT Ceramic Hydroxyapatite XT Media	Catalog #	Specification
Foresight CHT XT Plates	12003151	2 x 96-well, 20 µl
Foresight CHT XT RoboColumn Unit	12003152	200 µl
	12003148	600 µl
Foresight CHT XT Column	12003150	1 ml
	12003149	5 ml
CHT XT Media	12002457	10 g
	12002454	100 g
	12002456	1 kg
	12002455	5 kg

Note: Larger volumes and special packaging for industrial applications available upon request.

CHT Type I, Type II, and XT Media are available in prepacked GMP process-scale Foresight Pro Columns. A broad range of column diameters (5–33 cm inner diameter) and bed heights (10 or 20 cm) are available.

Visit [bio-rad.com/ForesightPro](https://www.bio-rad.com/ForesightPro) for more information.



Visit [bio-rad.com/MixedMode](https://www.bio-rad.com/MixedMode) for more information and to request samples.

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Nuvia aPrime 4A Resin is covered by U.S. Patent Number 9,669,402 and foreign counterparts.

BIO-RAD

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